Trigonometry: Trigonometric Identities in Degrees

Questions 1–54

Question 1. Write each of the following expressions in terms of $\sin x$:

a) $\sin(x + 180°)$
b) $\sin(-x)$
c) $\sin(x + 360°)$
d) $\sin(x - 180°)$
e) $\sin(x - 360°)$
f) $\sin(180° - x)$
g) $\sin(360° - x)$
h) $\sin(540° - x)$
i) $\sin(x - 540°)$
j) $\sin(x + 540°)$
k) $\sin(x + 720°)$
l) $\sin(900° - x)$
m) $\sin(3600° - x)$

Question 2. Write each of the following expressions in terms of $\cos x$:

a) $\cos(x - 180°)$
b) $\cos(-x)$
c) $\cos(180° - x)$
d) $\cos(x + 180°)$
e) $\cos(x + 360°)$
f) $\cos(360° - x)$
g) $\cos(x - 360°)$
h) $\cos(540° - x)$
i) $\cos(x + 540°)$
j) $\cos(x - 540°)$
k) $\cos(x - 720°)$
l) $\cos(900° - x)$
m) $\cos(3600° - x)$

Question 3. Write each of the following expressions in terms of $\tan x$:

a) $\tan(x + 180°)$
b) $\tan(-x)$
c) $\tan(x + 360°)$
d) $\tan(x - 180°)$
e) $\tan(360° - x)$
f) $\tan(180° - x)$
g) $\tan(x - 360°)$
h) $\tan(540° - x)$
i) $\tan(x - 540°)$
j) $\tan(x + 540°)$
k) $\tan(x + 720°)$
l) $\tan(900° - x)$
m) $\tan(3600° - x)$

Question 4. Write each of the following expressions in terms of $\cos x$:

a) $\sin(x + 90°)$
b) $\sin(x - 90°)$
c) $\sin(x + 270°)$
d) $\sin(90° - x)$
e) $\sin(270° - x)$
f) $\sin(x + 450°)$
g) $\sin(x - 450°)$
h) $\sin(450° - x)$
i) $\sin(x - 630°)$
j) $\sin(630° - x)$
k) $\sin(x + 810°)$
l) $\sin(990° - x)$
m) $\sin(9090° - x)$

Question 5. Write each of the following expressions in terms of $\sin x$:

a) $\cos(x + 90°)$
b) $\cos(x - 90°)$
c) $\cos(x + 270°)$
d) $\cos(90° - x)$
e) $\cos(270° - x)$
f) $\cos(x + 450°)$
g) $\cos(x - 450°)$
h) $\cos(450° - x)$
i) $\cos(x - 630°)$
j) $\cos(630° - x)$
k) $\cos(x + 810°)$
l) $\cos(990° - x)$
m) $\cos(9450° - x)$

Question 6. It is given that $\sin x = \frac{1}{3}$ and x is acute. Find the exact values of:

a) $\cos x$
b) $\tan x$

Question 7. It is given that $\cos x = \frac{2}{7}$ and x is acute. Find the exact values of:

a) $\sin x$

b) $\tan x$

Question 8. It is given that $\tan x = \frac{3}{10}$ and x is acute. Find the exact values of:

a) $\sin x$

b) $\cos x$

Question 9. It is given that $\sin x = \frac{2}{9}$ and x is obtuse. Find the exact values of:

a) $\cos x$

b) $\tan x$

Question 10. It is given that $\cos x = -\frac{3}{7}$ and x is obtuse. Find the exact values of:

a) $\sin x$

b) $\tan x$

Question 11. It is given that $\tan x = -\frac{11}{20}$ and x is obtuse. Find the exact values of:

a) $\sin x$

b) $\cos x$

Question 12. It is given that $\sin x = -\frac{5}{6}$ and $180° < x < 270°$. Find the exact values of:

a) $\cos x$

b) $\tan x$

Question 13. It is given that $\cos x = -\frac{1}{3}$ and $180° < x < 270°$. Find the exact values of:

a) $\sin x$

b) $\tan x$

Question 14. It is given that $\tan x = \frac{7}{2}$ and $180° < x < 270°$. Find the exact values of:

a) $\sin x$

b) $\cos x$

Question 15. It is given that $\sin x = -\frac{1}{10}$ and $270° < x < 360°$. Find the exact values of:

a) $\cos x$

b) $\tan x$

Question 16. It is given that $\cos x = \frac{14}{15}$ and $270° < x < 360°$. Find the exact values of:

a) $\sin x$

b) $\tan x$

Question 17. It is given that $\tan x = -\frac{20}{3}$ and $270° < x < 360°$. Find the exact values of:

a) $\sin x$

b) $\cos x$

Question 18. It is given that $\cos 75° = \frac{\sqrt{2-\sqrt{3}}}{2}$.

a) Find the value of $\sin 75°$

b) Using your answer to a), show that $\tan 75° = \sqrt{7+4\sqrt{3}}$

Question 19. Simplify the following trigonometric expressions as far as possible, writing your answers in terms of $\sin x$ and $\cos x$ only:

a) $\cos x \tan x$
b) $\sin x \tan x$
c) $\frac{\tan x}{\sin x}$
d) $\frac{\tan x}{\cos x}$

Question 20. Simplify the following trigonometric expressions as far as possible, writing your answers in terms of $\sin x$ and $\cos x$ only:

a) $\sin x \cos x \tan x$
b) $\sin x \cos^2 x \tan^2 x$
c) $\frac{\sin x}{\cos x \tan^2 x}$
d) $\frac{\tan^2 x}{\sin x \cos x}$

Question 21. Simplify the following trigonometric expressions:

a) $1 - \cos^2 x$

b) $\sqrt{1 - \sin^2 x}$

c) $9\cos^2 x - 9$

d) $8\cos^2 x + 8\sin^2 x$

Question 22. Simplify the following trigonometric expressions:

a) $\frac{1-\sin^2 x}{\cos^2 x}$

b) $\frac{1-\cos^2 x}{\sin^2 x}$

c) $\frac{4-4\cos^2 x}{5-5\sin^2 x}$

d) $\frac{\tan x}{6-\frac{6}{\cos^2 x}}$

Question 23. Solve the following equations:

a) $\sin\theta = \cos\theta$, $0° \leq \theta < 360°$

b) $2\sin\theta = 3\cos\theta$, $0° \leq \theta < 360°$

c) $8\sin\theta + 5\cos\theta = 0$, $0° \leq \theta < 360°$

d) $7\cos\theta - 2\sin\theta = 0$, $0° \leq \theta < 360°$

Question 24.

a) Sketch the graphs of $y = \cos x$ and $y = 2\sin x$ on the same axes between $0°$ and $360°$.

b) Determine the coordinates of where the two graphs intersect.

Question 25. Solve the following equations:

a) $6\sin 3\theta = 5\cos 3\theta$, $0° \leq \theta < 240°$

b) $7\sin \frac{1}{2}\theta - 9\cos \frac{1}{2}\theta = 0$, $0° \leq \theta < 720°$

c) $11\cos \frac{1}{3}\theta - 25\sin \frac{1}{3}\theta = 0$, $0° \leq \theta < 2160°$

Question 26. Solve the following equations:

a) $4\tan\theta + \cos\theta = 0$, $0° \leq \theta < 360°$

b) $\sin\theta\tan\theta + 6 = 0$, $0° \leq \theta < 360°$

c) $8\tan 2\theta + \cos 2\theta = 0$, $0° \leq \theta < 360°$

Question 27. Solve the following equations:

a) $2\cos^2\theta = 3\sin\theta$, $0° \leq \theta < 180°$

b) $4\sin^2\theta = 15\cos\theta$, $0° \leq \theta < 540°$

c) $7\cos^2\theta = 48\sin\theta$, $0° \leq \theta < 360°$

d) $10\sin^2\theta = 99\cos\theta$, $-180° \leq \theta < 360°$

Question 28. Write the expression $2\cos^4 x + \cos^2 x$ in terms of $\sin x$ only.

Question 29. Solve the following equations:

a) $2\cos^2\theta + 3\sin\theta - 3 = 0,\ 0° \leq \theta < 360°$

b) $3\sin^2\theta - 5\cos\theta - 5 = 0,\ 0° \leq \theta < 360°$

c) $6\cos^2\theta + \sin\theta - 4 = 0,\ 0° \leq \theta < 360°$

d) $20\sin^2\theta - 7\cos\theta - 14 = 0,\ 0° \leq \theta < 360°$

Question 30. Solve the following equations:

a) $3\sin^2\theta + \sin\theta - 3\cos^2\theta + 2 = 0,\ 0° \leq \theta < 360°$

b) $4\cos^2\theta - 16\cos\theta + 3\sin^2\theta - 19 = 0,\ 0° \leq \theta < 360°$

c) $10\sin^2\theta - 11\cos^2\theta + 46\sin\theta + 35 = 0,\ 0° \leq \theta < 360°$

Question 31.

a) Solve the equation $5\cos^2 x + 3\cos x + 2 = 4\sin^2 x$ for $-360° \leq x < 360°$

b) Hence find the smallest positive solution to the equation $5\cos^2(2\theta + 50°) + 3\cos(2\theta + 50°) + 2 = 4\sin^2(2\theta + 50°)$

Question 32. Solve $2\sin^2 x + 3 = \tan 5x - 2\cos^2 x$ for $0° \leq x < 72°$.

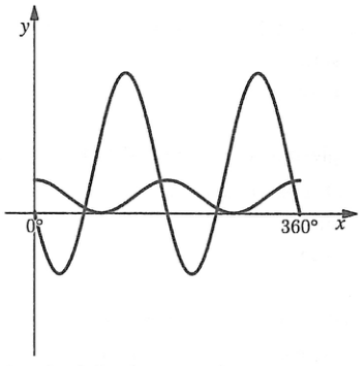
Question 33. Solve the following equations:

a) $10\sin\theta\tan\theta + 21 = 0,\ 0° \leq \theta < 360°$

b) $2\sin\theta\tan\theta + 20\cos\theta + 15 = 0,\ 0° \leq \theta < 360°$

c) $\frac{2\cos\theta}{\tan\theta} = 18\sin\theta - 3,\ 0° \leq \theta < 360°$

Question 34. The graph below shows the curves with equation $y = 12\sin^2 x - 28\sin x\cos x$ and $y = 5\cos^2 x$ from $0°$ to $360°$. Determine the coordinates of where the two graphs intersect.



Question 35. Solve the following equations:

a) $\sin\theta\tan\theta + 4\cos\theta = 5\sin\theta$, $0° \leq \theta < 360°$

b) $5\sin\theta\tan\theta = 3\cos\theta + 14\sin\theta$, $0° \leq \theta < 360°$

c) $10\sin\theta\tan\theta + 3\sin\theta = 18\cos\theta$, $0° \leq \theta < 360°$

Question 36. Solve the following equations:

a) $2\cos\theta + 13\cos\theta\tan\theta + 6\sin\theta\tan\theta = 0$, $0° \leq \theta < 360°$

b) $15\cos\theta + 6\sin\theta\tan\theta = 19\cos\theta\tan\theta$, $0° \leq \theta < 360°$

Question 37. Prove the following trigonometric identities:

a) $(\sin x + \cos x)^2 \equiv 1 + 2\sin x \cos x$

b) $\sin^3 x + \cos^3 x \equiv (\sin x + \cos x)(1 - \sin x \cos x)$

c) $1 + \tan^2 x \equiv \frac{1}{\cos^2 x}$

d) $\frac{1}{\tan^2 x} + 1 \equiv \frac{1}{\sin^2 x}$

Question 38. Prove the following trigonometric identities:

a) $\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} \equiv \frac{1}{\sin x \cos x}$

b) $\frac{1}{\cos x + 1} - \frac{1}{\cos x - 1} \equiv \frac{2}{\sin^2 x}$

c) $\frac{\cos x}{1 - \sin x} + \frac{\cos x}{1 + \sin x} \equiv \frac{2}{\cos x}$

d) $\frac{\cos x}{\sin x - 1} + \frac{\cos x}{\sin x + 1} \equiv -2\tan x$

Question 39. Prove the following trigonometric identities:

a) $\frac{1}{1+\tan^2 x} \equiv \cos^2 x$

b) $\frac{\tan x}{1+\tan^2 x} \equiv \sin x \cos x$

c) $\frac{1}{1-\tan^2 x} \equiv \frac{\cos^2 x}{1-2\sin^2 x}$

d) $\frac{\cos^2 x}{1-\tan^2 x} \equiv \frac{1}{1-\tan^4 x}$

Question 40. Prove the following trigonometric identities:

a) $\frac{1}{\frac{1}{\sin^2 x}-1} \equiv \tan^2 x$

b) $\frac{1}{\frac{1}{\cos^2 x}-1} \equiv \frac{1}{\sin^2 x} - 1$

c) $\frac{1}{\frac{1}{\tan^2 x}+1} \equiv \sin^2 x$

d) $\frac{1}{\frac{1}{\sin^2 x}+\frac{1}{\cos^2 x}} \equiv (\sin x \cos x)^2$

Question 41. Prove the following trigonometric identities:

a) $\sin^4 x - \cos^4 x \equiv \sin^2 x - \cos^2 x$

b) $\sin^2 x + \frac{\sin^4 x}{\cos^2 x} \equiv \tan^2 x$

c) $\cos^4 x + \sin^2 x + \sin^2 x \cos^2 x + \sin^2 x \tan^2 x \equiv \cos^2 x + \tan^2 x$

d) $\sin^2 x \cos^2 x - 1 + \sin^4 x - \tan^2 x \equiv \sin^2 x - \frac{1}{\cos^2 x}$

Question 42. Prove the following trigonometric identities:

a) $\frac{\sin x}{1-\tan x} + \frac{\cos x}{1-\frac{1}{\tan x}} \equiv 0$

b) $\frac{\cos x}{1-\tan x} + \frac{\sin x}{1-\frac{1}{\tan x}} \equiv \sin x + \cos x$

c) $\frac{\sin x}{1-\tan x} - \frac{\cos x}{1-\frac{1}{\tan x}} \equiv \frac{2\sin x - 2\cos x}{2-\frac{1}{\sin x \cos x}}$

d) $\frac{\cos x}{1-\tan x} - \frac{\sin x}{1-\frac{1}{\tan x}} \equiv \frac{1}{\cos x - \sin x}$

Question 43. Prove the following trigonometric identities:

a) $\frac{\sin x}{1-\cos x} \equiv \frac{1+\cos x}{\sin x}$

b) $\frac{\cos x}{1-\sin x} \equiv \frac{1+\sin x}{\cos x}$

c) $\frac{\sin x}{1-\tan x} \equiv \frac{\sin x \cos x}{\cos x-\sin x}$

d) $\frac{1-\sin x}{1-\cos x} \equiv \frac{1+\cos x}{\sin^2 x} - \frac{1+\cos x}{\sin x}$

Question 44. Prove the following trigonometric identities:

a) $\frac{\tan x+\frac{1}{\tan x}}{\frac{1}{\cos x}+\frac{1}{\sin x}} \equiv \frac{1}{\sin x+\cos x}$

b) $\frac{\cos x-\frac{1}{\cos x}}{\frac{1}{\sin x}-\frac{1}{\cos x}} \equiv \frac{\sin^2 x}{1-\frac{1}{\tan x}}$

c) $\frac{\sin x-\frac{1}{\sin x}}{\frac{1}{\sin x}-\frac{1}{\cos x}} \equiv \frac{\cos^3 x}{\sin x-\cos x}$

Question 45. Prove the following trigonometric identities:

a) $\frac{\sin x \cos x}{1+\sin x+\cos x-\cos^2 x} \equiv \frac{1}{\frac{1}{\sin x}+\frac{1}{\cos x}+\tan x}$

b) $\frac{\sin x \cos^2 x}{\cos^3 x-2\cos^2 x-\cos x+1} \equiv \frac{1}{\frac{\tan x}{\cos x}-\frac{1}{\sin x}-\tan x}$

Question 46. Prove the following trigonometric identities:

a) $(\frac{1}{\cos x}-\sin x)(\frac{1}{\sin x}-\cos x) \equiv \frac{(\sin x \cos x-1)^2}{\sin x \cos x}$

b) $(\frac{1}{\sin x}-\sin x)(\frac{1}{\cos x}-\cos x) \equiv \frac{1}{\tan x+\frac{1}{\tan x}}$

Question 47. Prove the following trigonometric identity: $\frac{2\cos^2 x-1}{\cos x-\sin x} \equiv \frac{1}{\sin x-\cos x+\frac{2\cos^2 x}{\sin x+\cos x}}$.

Question 48. Prove the following trigonometric identity: $\sin^2 A+\sin^2 B-1 \equiv \sin^2 A\sin^2 B-\cos^2 A\cos^2 B$.

Question 49. Isla is attempting to solve the equation $\sqrt{2}\cos^2 x + \sin x - \sqrt{2} = 0$ for $180° < x < 360°$. Her attempt is shown below:

$$\begin{aligned}
\sqrt{2}\cos^2 x + \sin x + \sqrt{2} &= 0 \\
\Rightarrow \sqrt{2}(\sin^2 x - 1) + \sin x + \sqrt{2} &= 0 \\
\Rightarrow \sqrt{2}\sin^2 x - \sqrt{2} + \sin x + \sqrt{2} &= 0 \\
\Rightarrow \sqrt{2}\sin^2 x + \sin x &= 0 \\
\Rightarrow \sqrt{2}\sin x + 1 &= 0 \\
\Rightarrow \sin x &= -\frac{1}{\sqrt{2}} \\
\Rightarrow x &= -45°
\end{aligned}$$

Explain the two errors that Isla has made.

Question 50.

a) Show that the equation $20\sin x \tan x + 31\sin x = 7\cos x$ can be written in the form $(4\tan x + 7)(5\tan x - 1) = 0$

b) Hence solve the equation $20\sin 4\theta \tan 4\theta + 31\sin 4\theta = 7\cos 4\theta$ for $0° \leq \theta < 45°$

Question 51.

a) Show that $\frac{\tan x}{\frac{1}{\cos x}-\cos x} \equiv \frac{1}{\sin x}$

b) Using your answer to a), solve the equation $\frac{\tan x}{\frac{1}{\cos x}-\cos x} = 6$ for $0° \leq x < 360°$

c) Hence solve the equation $\frac{\tan(2\theta-55°)}{\frac{1}{\cos(2\theta-55°)}-\cos(2\theta-55°)} = 6$ for $0° \leq \theta < 180°$

Question 52.

a) Show that $\frac{\cos^3 x}{\sin x} + \sin x \cos x \equiv \frac{1}{\tan x}$

b) Using your answer to a), solve the equation $\frac{\cos^3 x}{\sin x} + \sin x \cos x = \frac{1}{20}$ for $0° \leq x < 360°$

c) Hence solve the equation $\frac{\cos^3(2\theta+9°)}{\sin(2\theta+9°)} + \sin(2\theta+9°)\cos(2\theta+9°) = \frac{1}{20}$ for $0° \leq \theta < 180°$

Question 53. Solve the equation $\sin^4\theta - \cos^4\theta = \frac{1}{2}$ for $0° \leq \theta \leq 360°$.

Question 54. Solve the equation $2\cos^4\theta + \cos^2\theta\sin\theta - 2\cos^2\theta = 0$ for $0° \leq \theta \leq 360°$.